

MAKING VISIONS WORK.

Digital Businesses, Digital Experiences & Products



Software Requirement Engineering

bbv Software Services AG | Nov, 2024



consimo
fa:r Parifonds Bau

Presentation title with customer logo

bbv Software Services AG | Date

Agenda

1. **Welcome & introduction of bbv**
2. **Introduction to Requirement Engineering**
3. **Roles of Dev, QC in Requirement Engineering**
4. **Understanding Functional Requirement**
Elicitation, analysis, specification
5. **Top-down approach and prototyping (C4)**
(e.g., Figma, Balsamiq, Sketch, visily)
6. **Non-functional requirements**
QA
7. **Writing Clear Requirements**
8. **Q&A**



Workshop

Requirement Engineering Foundation

Join our Requirement Engineering Workshop to learn essential skills for developers and QC professionals, covering roles, requirements, tools, and best practices to enhance collaboration and improve software product quality.

 Speakers:

Dao Vo - Chief Technology Officer, bbv Vietnam

 Date: 01 November 2024

 Time: 10:00 - 11:30



Software Requirement?

What's Software Requirement?

- Definition: **Description and Specifications of a system**
- Topics covered:
 - Functional and Non-functional requirement
 - User Requirements
 - System requirements
 - The software requirements document

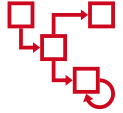


What is a requirement?

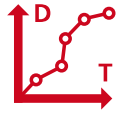
- It may range from a high-level abstract statement of a service or of a system constraint to a detailed mathematical functional specification
- This is inevitable as requirements may serve a dual function
 - May be the basis for a bid for a contract - therefore must be open to interpretation
 - May be the basis for the contract itself - therefore must be defined in detail
 - Both these statements may be called requirements



Business User Expectation



Transition to modern technologies



Further development without limits



Accelerate time-to-market



100 times more quality and faster

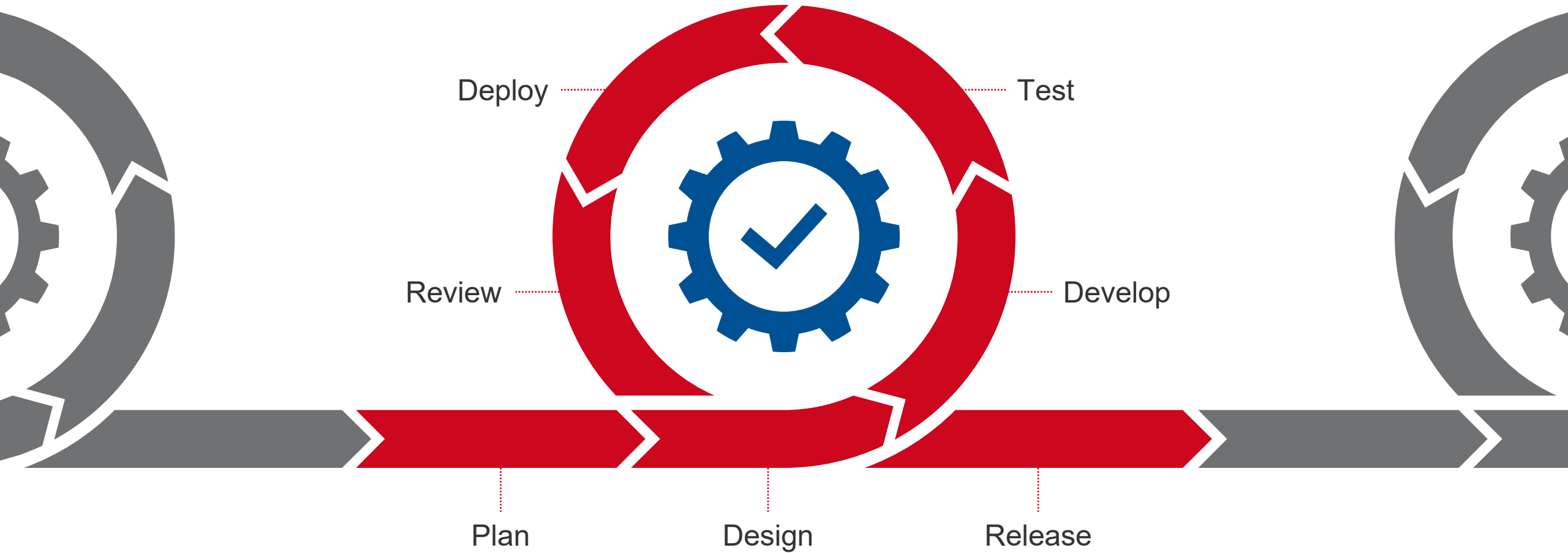


Improve user experience

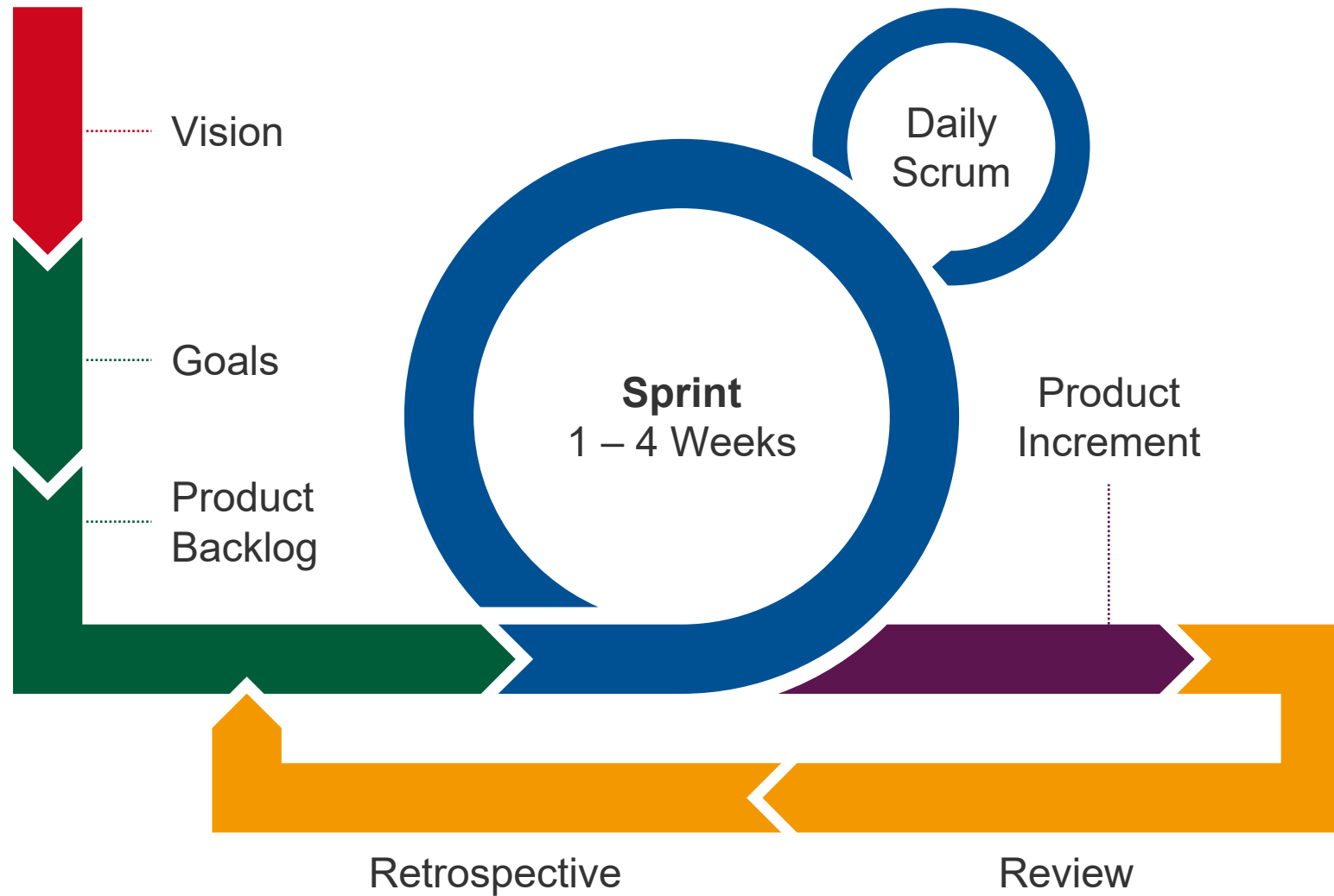


Product development procedure

Stage	Fundamentals	Exploration	Pilot	Product
Duration	1–2 days	1–2 months	3–9 months	Years
Activities	• Thematic introduction • Challenge workshop	• Service Design • Prototyping, UX • Architecture workshop	• Lean Development	• Agile Development
Roles	• Consultant • RE / UX • PL • ML / AM • Architect / Eng. / QS	• Consultant • RE / UX • PL • ML / AM • Architect / Eng. / QS	• Consultant • RE / UX • PL • ML / AM • Architect / Eng. / QS	• Consultant • RE / UX • PL • ML / AM • Architect / Eng. / QS
Results	• Product vision • Common understanding • Personas & system context • Requirements & risk management	• α-customer Feedback • Value proposition & business model • PoC (internal expertise) • Architecture blueprint & technology stack • Procedure & (team) setup	• β-customer Feedback • Minimal marketable product • Implementation organisation	• Continuous business elaboration • Portfolio management & service organisation (DevOps)



Scrum process

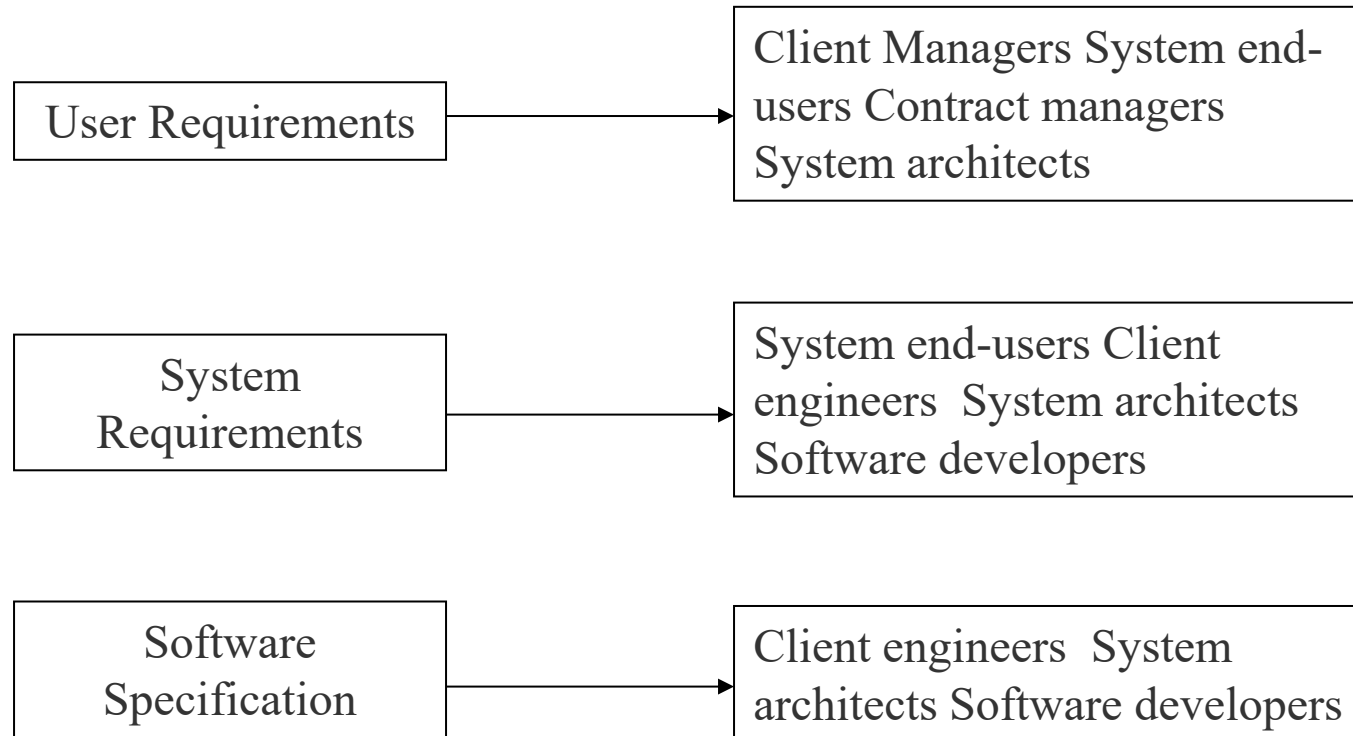


Types of requirements

- User requirements
 - Statements in natural language (NL) plus diagrams of the services the system provides and its operational constraints. Written for customers
- System requirements
 - A structured document setting out detailed descriptions of the system services. Written as a contract between client and contractor
- Software specification
 - A detailed software description which can serve as a basis for a design or implementation. Written for developers



Requirements Targets



Requirements Types

1. Functional requirements: services the system should provide
2. Non-functional requirements: constraints on the services of functions offered by the system. e.g. speed, time to market
3. Domain requirements: related to the application domain of the system (may be functional or non-functional requirements)



Functional requirements

- Functionality or services that the system is expected to provide.
- Functional requirements may also explicitly state what the system shouldn't do.
- Functional requirements specification should be:
 - Complete: All services required by the user should be defined
 - Consistent: should not have contradictory definition (also avoid ambiguity → don't leave room for different interpretations)



Non-Functional requirements

- Requirements that are not directly concerned with the specific functions delivered by the system
- Typically relate to the system as a whole rather than the individual system features
- Often could be deciding factor on the survival of the system (e.g. reliability, cost, response time)



Goal of the project or summary of the customer benefit (e.g. significantly increase efficiency with a new software architecture and improved usability). → Quality Attributes

Software Development Quality Map

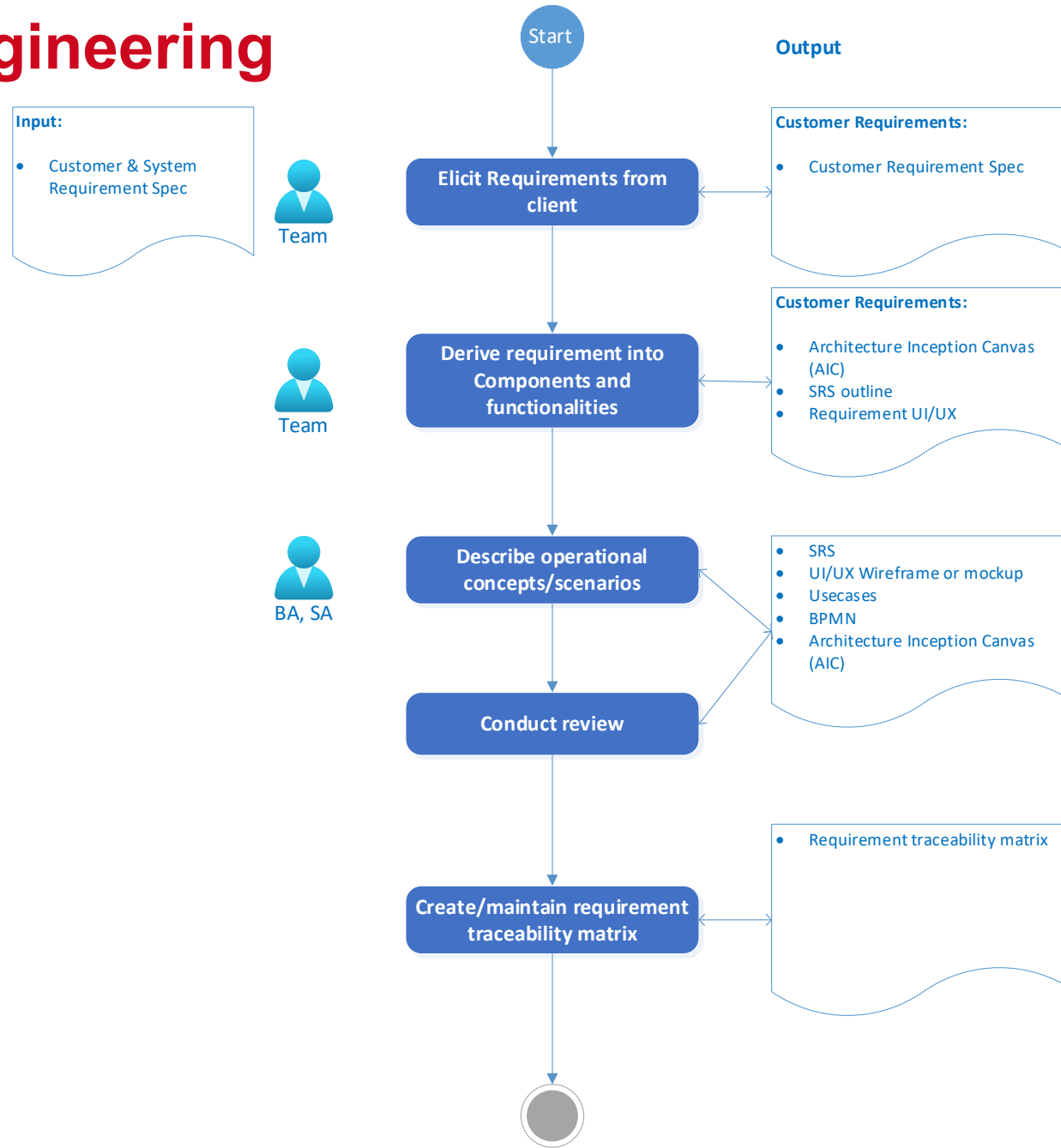


Cafe?

bbv



Requirement Engineering Process



Convert From Business Needs to Solution?



Working structure

«Overview»

Get an overview of the technology or concept; how to use it successfully.

«Exploration»

Develop promising ideas; identify potentials; create idea ranking list.

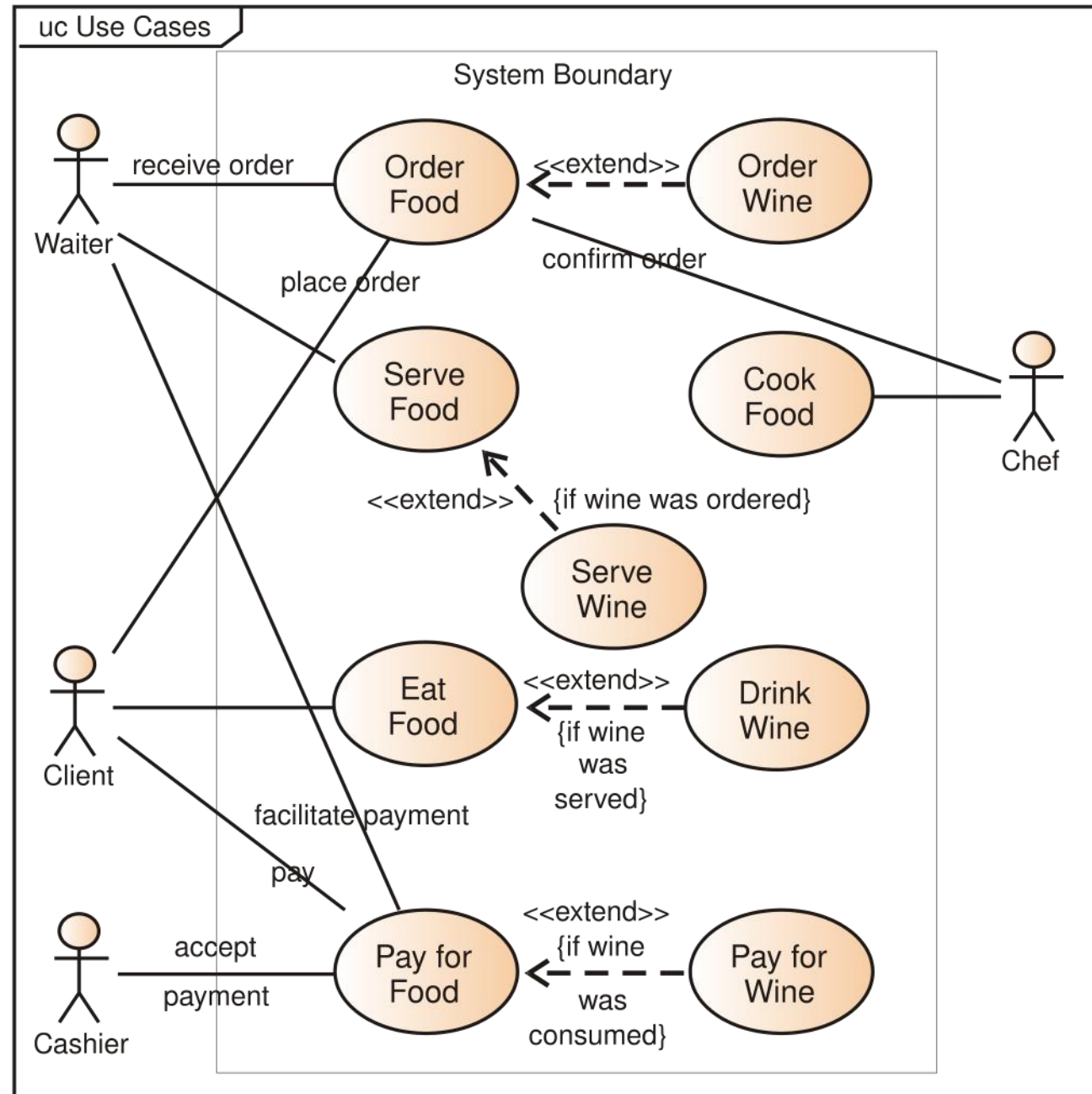
«Consolidation»

Concretize the best ideas; develop business cases.

Usecase Diagram

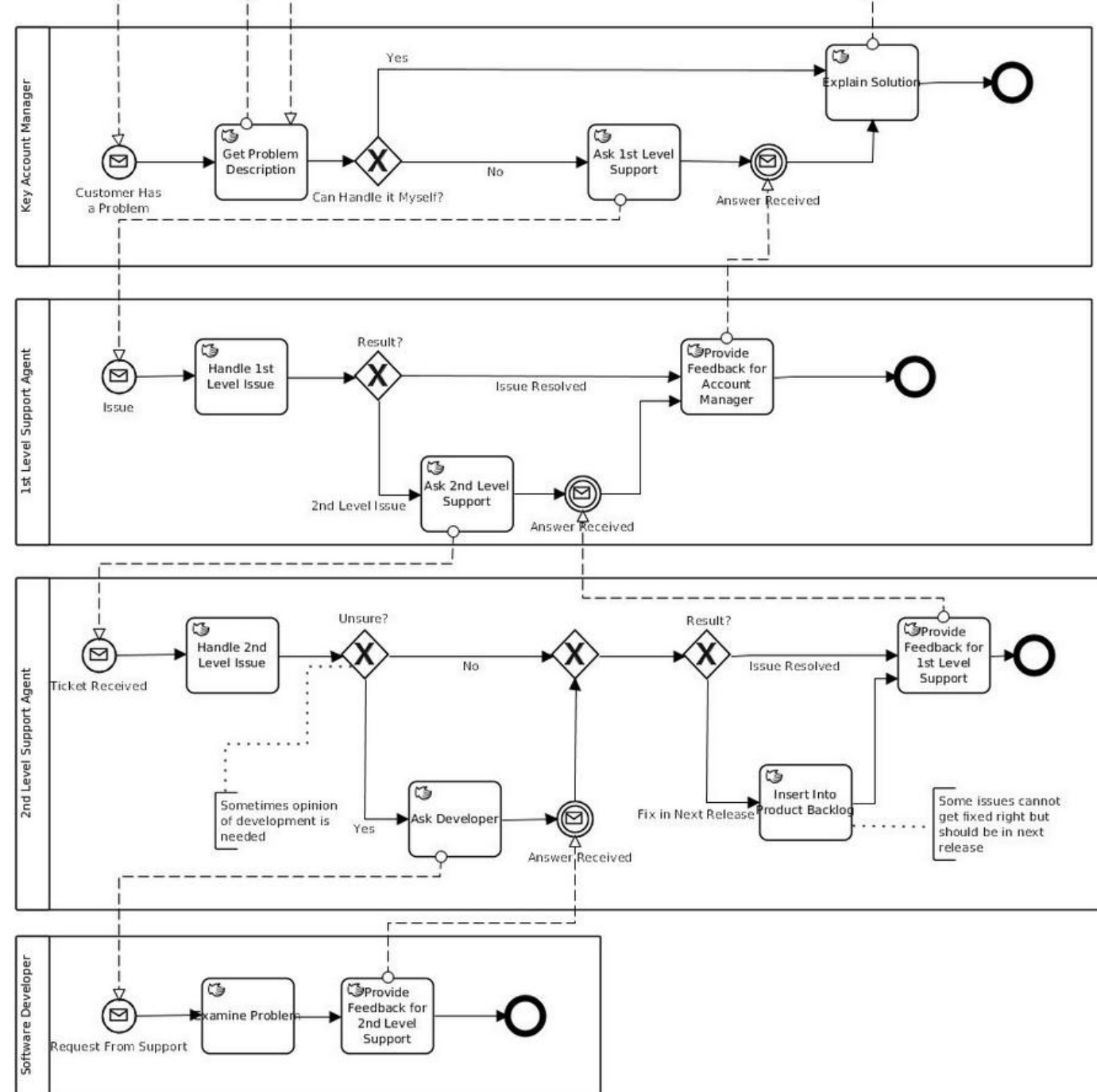
What are user personas?

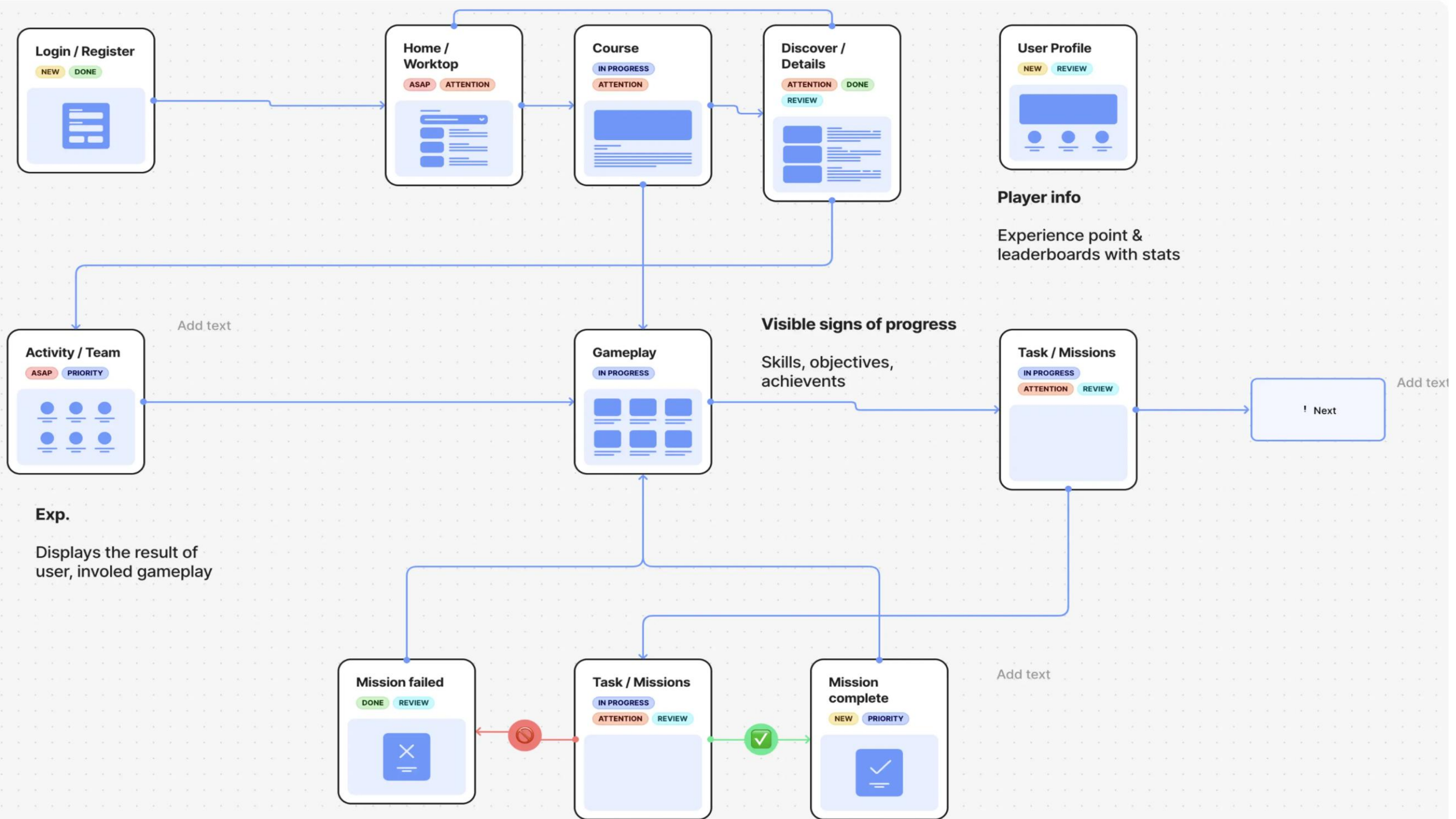
- Actors
- Functionals



Usecase Diagram

How to manage business flow between user personas



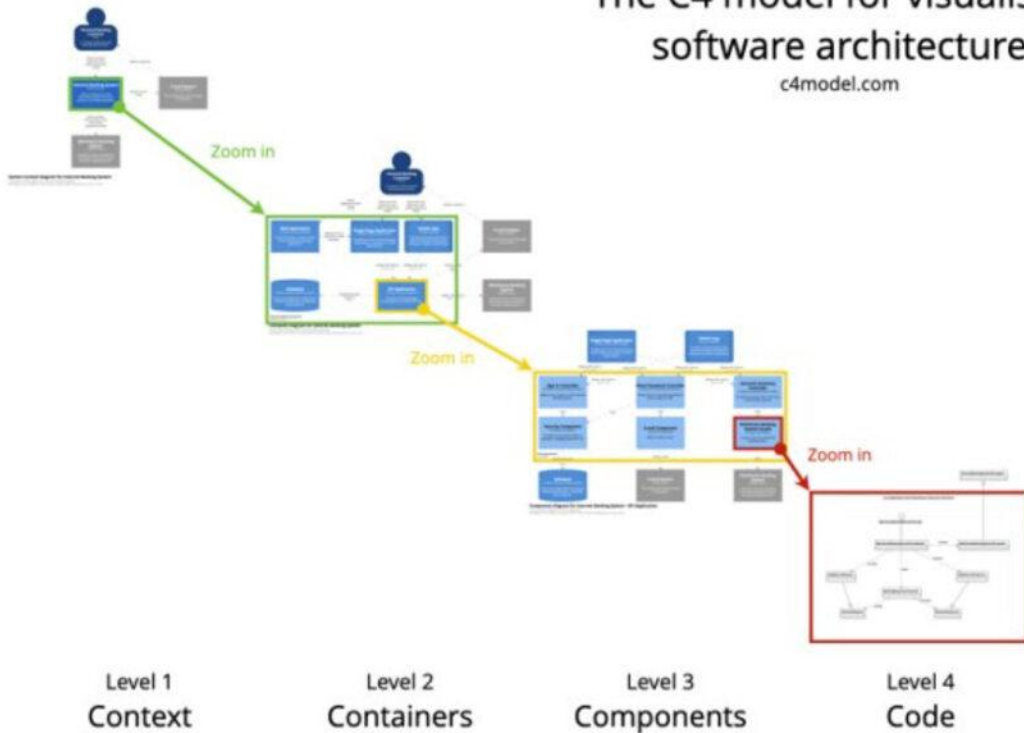


C4 Model

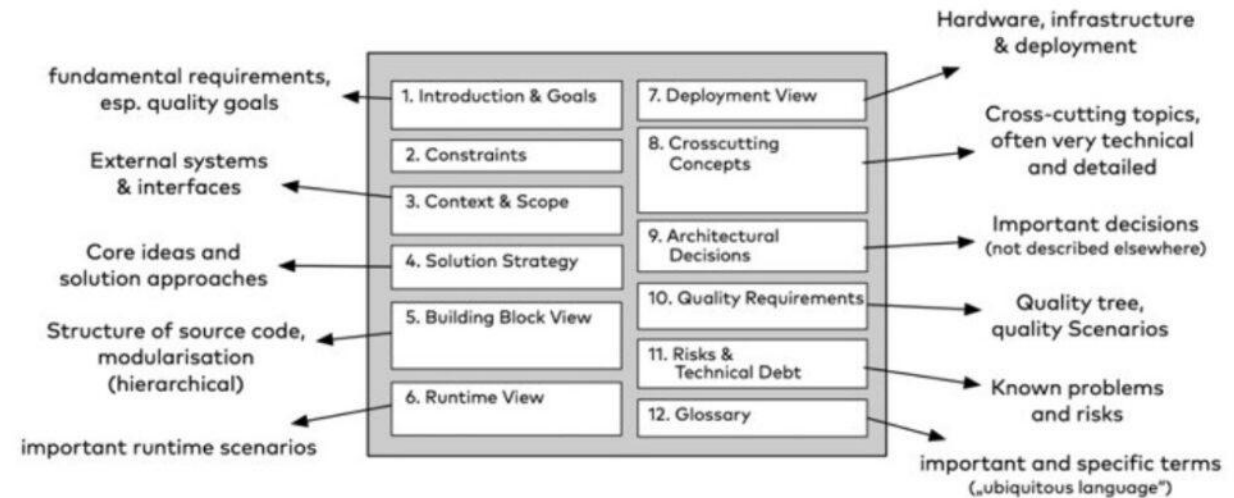
C4

The C4 model for visualising
software architecture

c4model.com

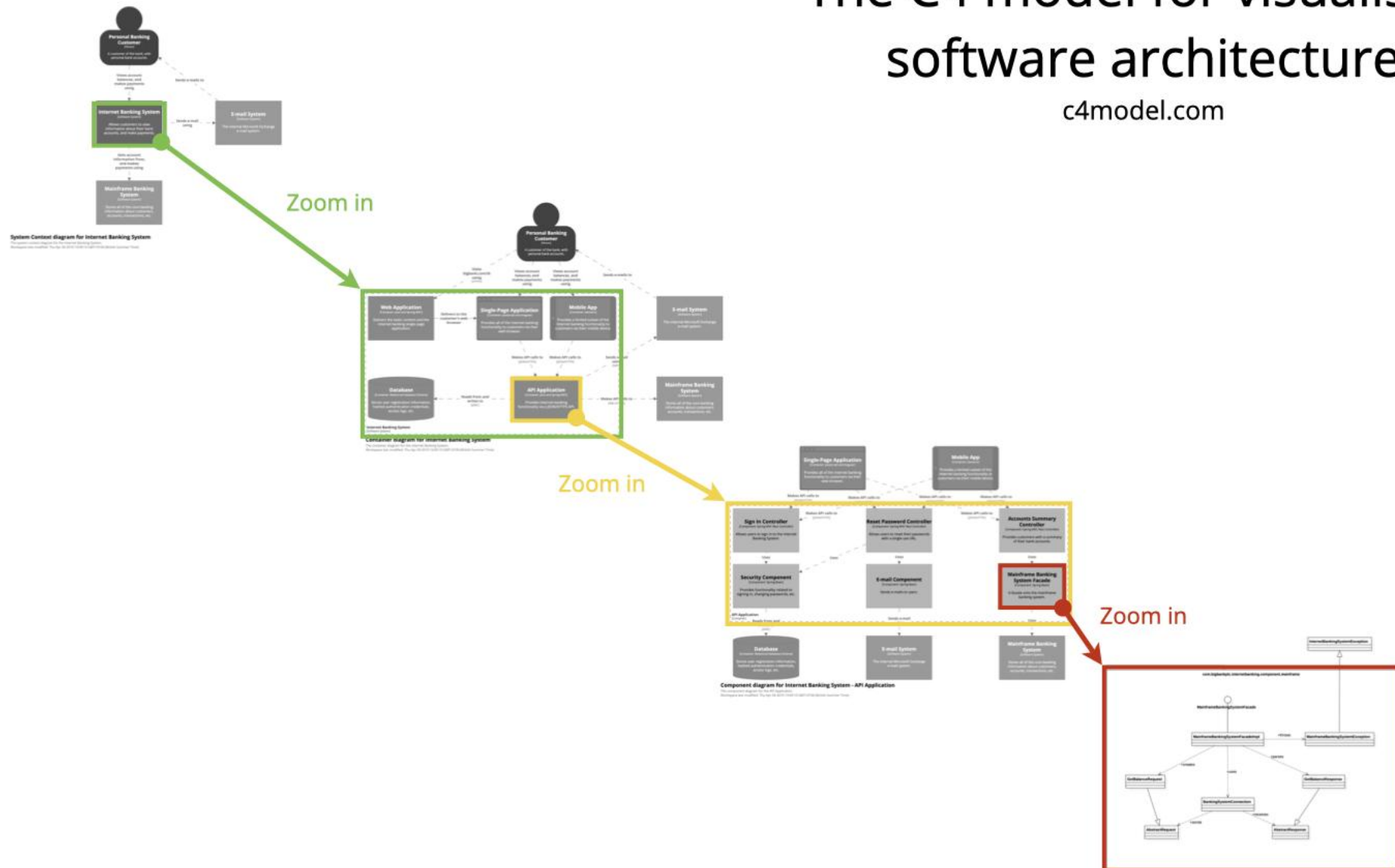


Arc42



The C4 model for visualising software architecture

c4model.com



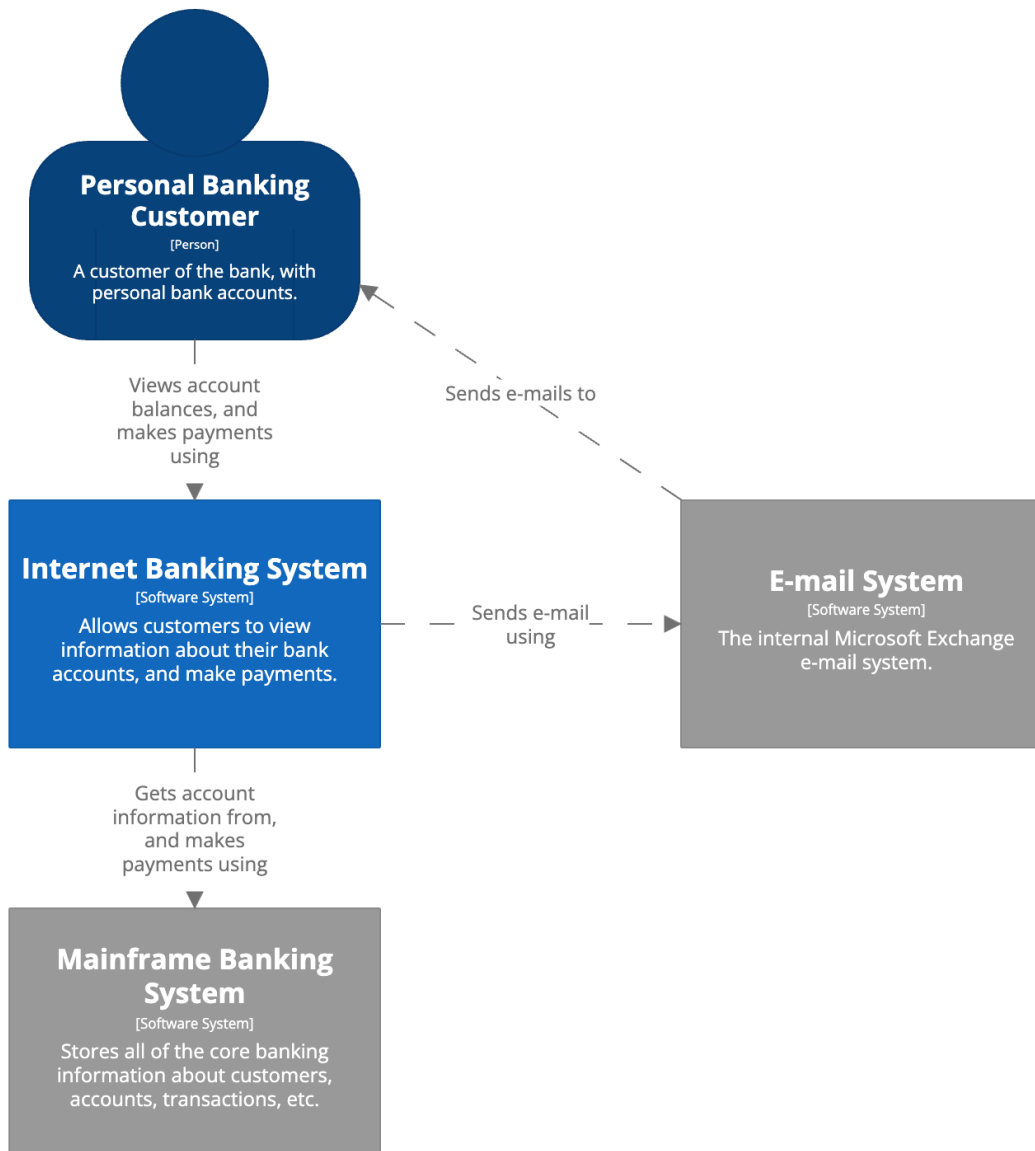
Level 1
Context

Level 2
Containers

Level 3
Components

Level 4
Code

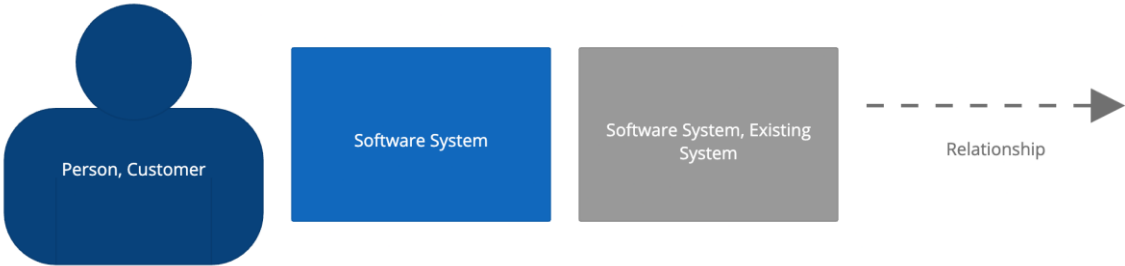
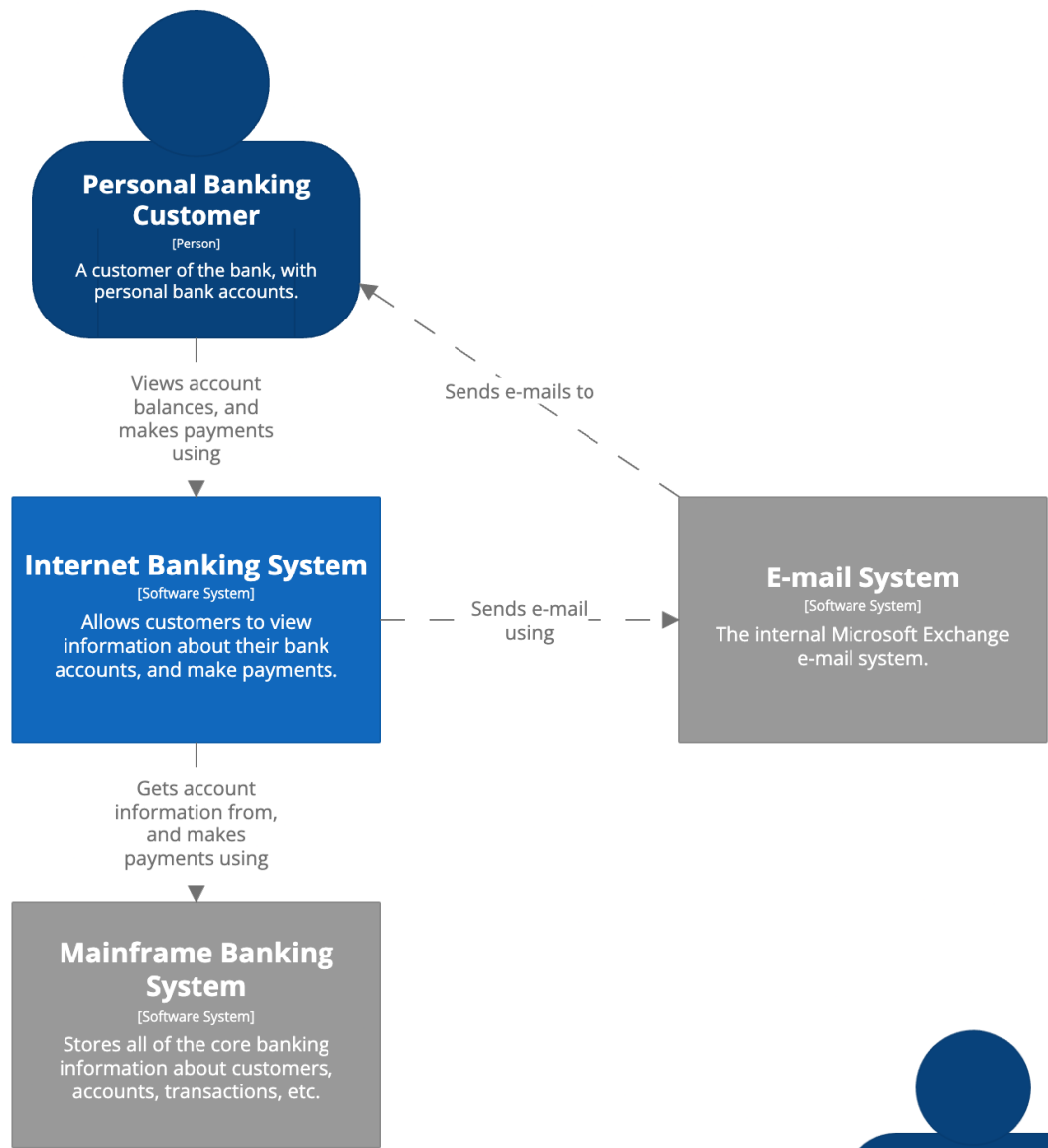
C1: Context



[System Context] Internet Banking System

The system context diagram for the Internet Banking System - diagram created with Structurizr.
Saturday, 11 November 2023 at 09:04 Greenwich Mean Time

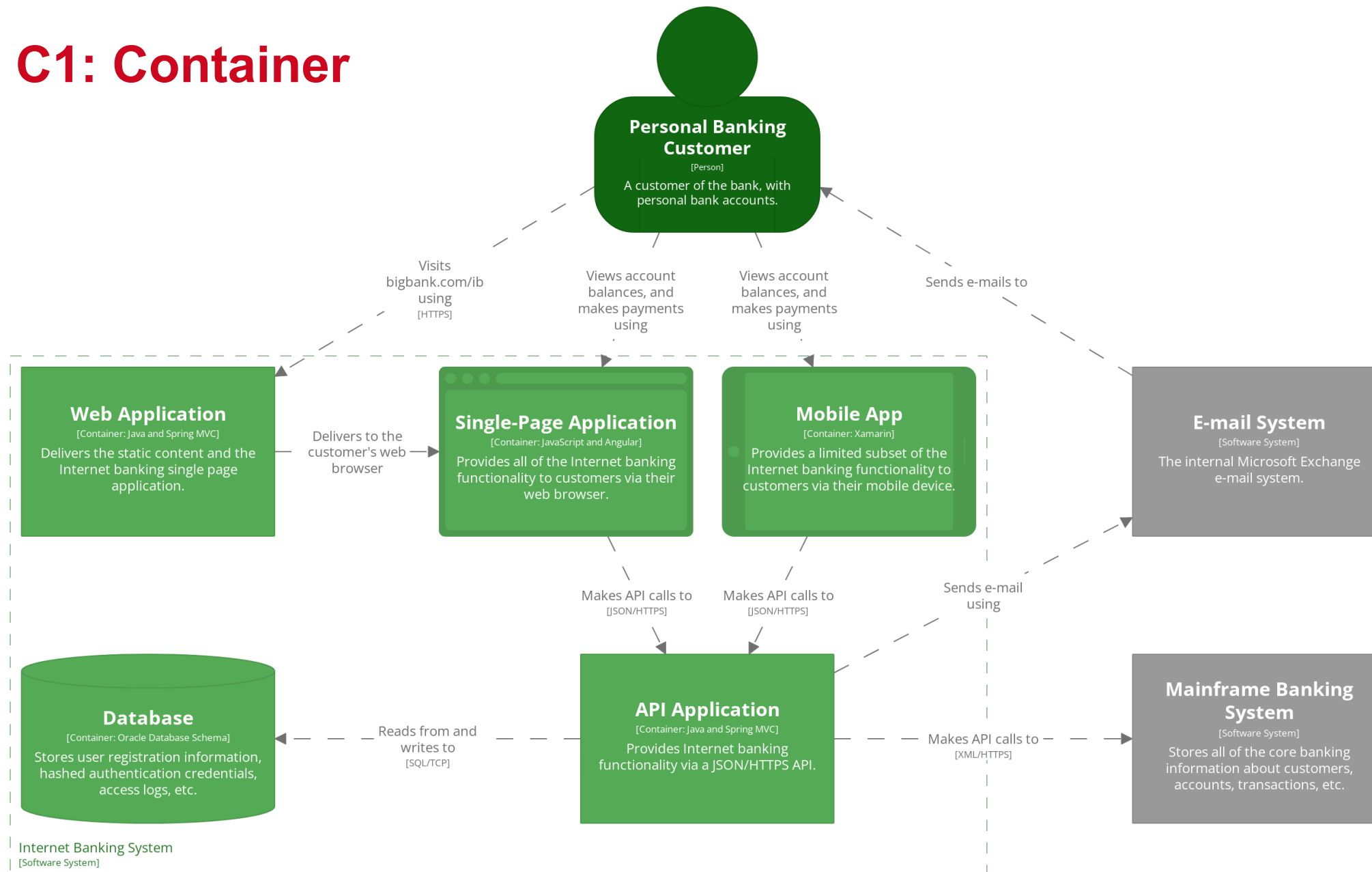
C1: Context



[System Context] Internet Banking System

The system context diagram for the Internet Banking System - diagram created with Structurizr.
Saturday, 11 November 2023 at 09:04 Greenwich Mean Time

C1: Container



[Container] Internet Banking System

The container diagram for the Internet Banking System - diagram created with Structurizr.

Wednesday, March 22, 2023 at 8:16 AM Coordinated Universal Time

Architecture Inception Canvas

Software System:

Designed by Team:

Workshop Date:

Iteration:



Business Case

Brief description of the business case or economic driver behind the software system.



Functional Overview

The most important functional requirements at a high level.



Business Context

Separate your system under construction as a black box from all its communication partners. Communication partners are neighbouring external systems and users.



Organisational Constraints

Any organisational requirement that limits the software architects freedom of decision.



Quality Goals

The three most important quality goals for the architecture, which have the highest priority for the most important stakeholder.



Technical Constraints

Any technical requirement that limits the software architects freedom of decision.



Architectural hypotheses

Resulting architectural hypotheses and important, expensive, large-scale or risky architectural decisions, including justifications.



Technical Challenges & Risks

Identified current known challenges and technical risks





Q&A